App Switching and Mental Overload Throughout the Day

Introduction

This project explores how app switching behavior and notifications relate to mental overload throughout the day. The main goal was to understand whether increased screen activity leads to higher mental strain, especially later in the day compared to the morning.

Data Collection Process and Rationale

Data was collected daily from February to March 2026 and continued until the end of the term. Each day, I recorded the number of app switches, the number of notifications received, and my mental overload at two different times of the day (morning and night).

Mental overload was self-reported on a scale from 1 to 5, where 1 represents very low stress and 5 represents very high stress. I chose this method because it allowed me to consistently track changes in my mental state throughout the day.

The reason for collecting this data was to better understand how frequent digital interactions, such as switching between apps and receiving notifications, may affect mental well-being. By collecting data over multiple weeks, I was able to observe patterns and trends rather than relying on a few isolated days.

Dataset

The dataset includes the following variables:

- Date
- App switches
- Total screen time
- Notifications
- Mental overload (morning)
- Mental overload (night)
- Daily average overload

Even though not all variables were used directly in the final visualization (such as total screen time and daily average overload), they were still collected and considered during the design process.

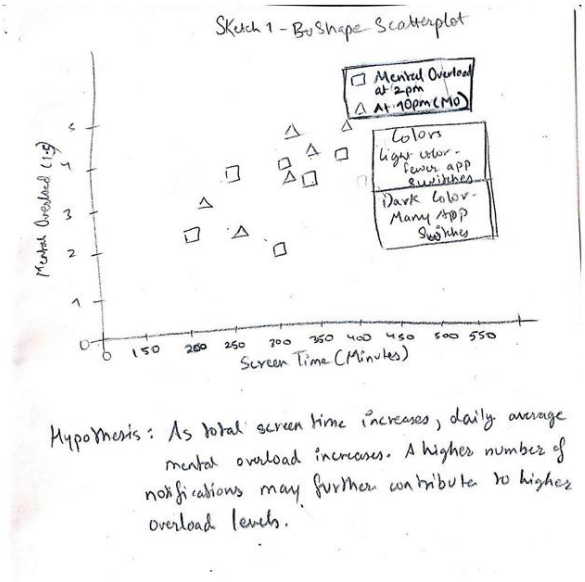
Date	App Switches	Total Screen Time (min)	Notification	Mental Overload (Morning)	Mental Overload (Night)	Daily Average Overload
Feb 8, 2026	110	420	89	3	4	3.5
Feb 9, 2026	142	510	118	4	5	4.5
Feb 10, 2026	156	545	132	4	5	4.5
Feb 11, 2026	128	470	104	3	5	4
Feb 12, 2026	135	490	115	4	4	4
Feb 13, 2026	85	210	70	2	3	2.5
Feb 14, 2026	60	150	45	1	2	1.5
Feb 15, 2026	75	185	60	2	3	2.5
Feb 16, 2026	95	310	85	3	4	3.5
Feb 17, 2026	148	525	125	4	5	4.5
Feb 18, 2026	90	300	80	3	4	3.5
Feb 19, 2026	105	360	95	3	4	3.5
Feb 20, 2026	120	420	105	4	4	4
Feb 21, 2026	130	460	112	4	5	4.5
Feb 22, 2026	145	500	120	4	5	4.5
Feb 23, 2026	155	540	130	5	5	5
Feb 24, 2026	165	580	145	5	5	5
Feb 25, 2026	140	500	120	4	5	4.5
Feb 26, 2026	125	450	110	4	4	4
Feb 27, 2026	100	320	90	3	4	3.5
Feb 28, 2026	86	260	75	2	3	2.5
Mar 1, 2026	70	200	60	2	2	2

Mar 2, 2026	90	310	85	3	3	3
Mar 3, 2026	115	420	100	4	4	4
Mar 4, 2026	135	480	115	4	5	4.5
Mar 5, 2026	153	520	130	5	5	5
Mar 6, 2026	120	400	105	4	4	4
Mar 7, 2026	83	270	80	3	3	3
Mar 8, 2026	65	180	55	1	2	1.5
Mar 9, 2026	98	330	90	3	4	3.5
Mar 10, 2026	135	510	120	4	5	4.5
Mar 11, 2026	110	550	135	5	5	5
Mar 12, 2026	75	380	100	3	4	3.5
Mar 13, 2026	67	21	65	2	3	2.5
Mar 14, 2026	60	160	53	1	2	1.5
Mar 15, 2026	130	470	110	4	5	4.5

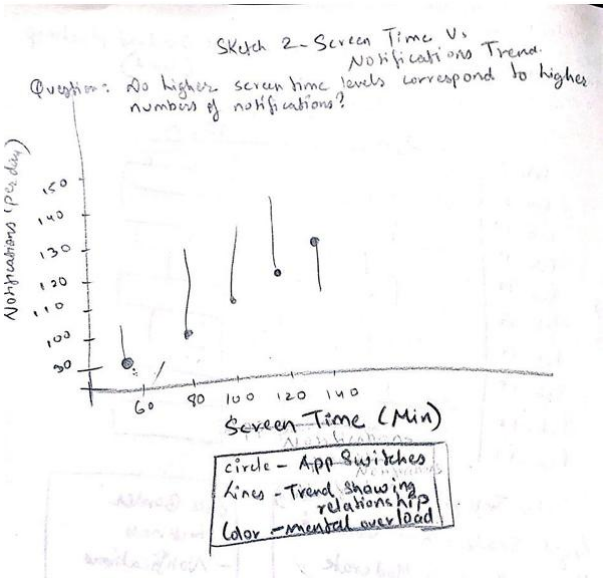
Alternative Visualizations and Comparison

During the design process, I explored multiple visualization ideas before choosing the final design.

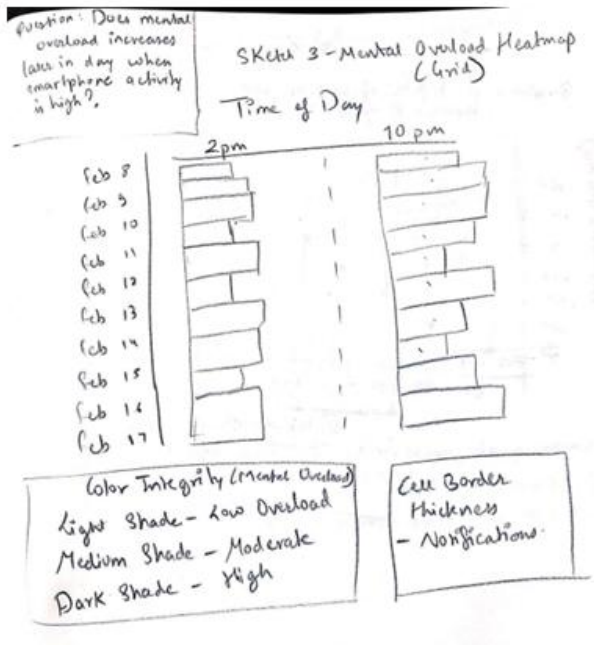
The first idea was a scatterplot showing screen time versus mental overload. This made it easy to see general trends, but it did not clearly compare morning and night values for each day.



The second idea was a trend-based visualization showing relationships between screen time and notifications using circles and color. While it showed multiple variables, it became visually complex and harder to interpret.



The final design uses a bar-based comparison with side-by-side rectangles for morning and night. This approach made it easier to compare changes throughout the day while still representing multiple variables clearly.



Changes During Process

During the data collection process, additional entries were added over time to extend the dataset until the end of the term. The structure of the data remained consistent, but more data points were included to improve accuracy and allow for better pattern observation. I also made small adjustments to the visualization design during development to make it clearer and easier to understand, such as refining the layout and improving how variables were represented.

Rationale for Final Visualization

The final visualization uses horizontal rectangles to represent daily data, which allows for easy comparison across time. Position is used to separate morning and night values, making differences between the two time periods clear.

Color represents mental overload, using a gradient from light to dark red to indicate increasing levels of stress. Rectangle width represents app switching activity, showing how frequently apps were used. Stroke thickness represents notification levels, adding another layer of information without cluttering the design.

This combination of visual encodings allows multiple variables to be displayed in a clear and organized way. Compared to other designs, this approach provides the best balance between readability and detail. According to Data to Viz, choosing appropriate visual encodings such as color and size improves clarity and interpretation, which influenced the design of this visualization.

Conclusions Based on Data

Based on the visualization, mental overload generally increases later in the day, especially on days with higher app switching activity. Darker colors appear more frequently in the night data, suggesting that prolonged screen use and frequent interruptions may contribute to increased mental strain.

Overall, the data supports the idea that higher digital activity is associated with higher mental overload, particularly in the evening.

Past and Future Data Collection Thoughts

One limitation of this dataset is that it only includes a few variables related to digital activity. In the future, I would collect additional data such as sleep quality, workload, or mood to better understand other factors that may influence mental overload.

If I were to continue this project, I would also collect data over a longer period of time to see if the patterns remain consistent.

Supporting Sources / References

D3.js Documentation. https://d3js.org

Data to Viz. https://www.data-to-viz.com

Course lecture materials and examples.